PCT/PCM 12: Guess What the Loss Prefers

4. Interpret the Figure

On the outlier dataset, the L2 curve reaches its minimum at the mean (approximately 16.6), while the L1 curve reaches its minimum at the median (approximately 10.5). The right panel shows that MSE prefers the blurred image (lower MSE = 189), whereas SSIM prefers the sharp image with +16 brightness (higher SSIM = 0.951).

5. Calculation

Given clean values {5, 6, 6, 7, 31}:

L2-optimal output = mean = (5 + 6 + 6 + 7 + 31) / 5 = 55 / 5 = 11.
L1-optimal output = median = 6.

The mean-based output is softer because the outlier (31) pulls the mean away from the majority of values, encouraging smoother predictions.

6. Prediction Card

(a) Blurred vs. +16-brightness copy:
MSE prefers: Blurred.
SSIM prefers: +16 brightness.

(b) One dead pixel vs. global haze:
MSE prefers: One dead pixel.
SSIM prefers: One dead pixel.

This prediction assumes the dead pixel is isolated and the haze affects a substantial part of the image. The exact MSE result depends on the severity of each distortion.